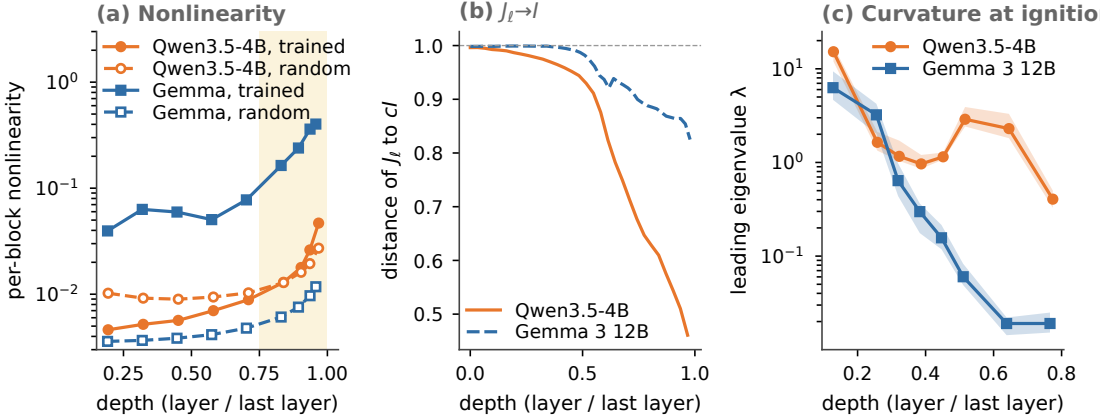
(a) Nonlinearity
Qwen3.5-4B, trained
Qwen3.5-4B, random
Gemma, trained
Gemma, random
per-block nonlinearity
10^0
10^{-1}
10^{-2}
0.25
0.50
0.75
1.00
depth (layer / last layer)
(b) $J_\ell \to I$
distance of J_ℓ to cI
1.0
0.9
0.8
0.7
0.6
0.5
Qwen3.5-4B
Gemma 3 12B
0.0
0.5
1.0
depth (layer / last layer)
(c) Curvature at ignition
Qwen3.5-4B
Gemma 3 12B
leading eigenvalue λ
10^1
10^0
10^{-1}
0.2
0.4
0.6
0.8
depth (layer / last layer)